PRICE LEVEL DETERMINATION IN A MONETARY ECONOMY

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PLAN

- We study how the price level (and inflation) is determined in a monetary economy.
- This will later serve us as a framework to study interactions of fiscal and monetary policy.
- We will start with a very simple environment and then gradually add more features.

ENDOWMENT ECONOMY

ENDOWMENT ECONOMY

- Every period there is an endowment of nonstorable goods y_t.
- For simplicity assume $y_t = y$ for all t.
- Let P_t be the price level in period t: one unit of goods costs P_t units of account.
- There is a representative household that trades contingent claims (Arrow securities).
- The household maximizes

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t u(c_t).$$

- No government purchases of goods.

ASSET PRICES

- Let $q_{t,t+1}$ be the real price (in goods) of a claim that pays one unit of goods in period $t+1$ in a particular state of the world.
- The nominal contingent claim price is

$$Q_{t,t+1} = q_{t,t+1} \frac{P_t}{P_{t+1}}.$$

- The nominal interest rate i_t satisfies

$$\frac{1}{1+i_t} = \mathbb{E}_t \, Q_{t,t+1}.$$

ASSET PRICES

- In this economy $c_t = y$.
- The (real) SDF, $\beta u'(c_t + 1)/u'(c_t)$, is constant and equal to β.
- The real interest rate r is also constant and satisfies

$$\frac{1}{1+r} = \beta.$$

- The nominal discount factor is $Q_{t,t+1} = \beta \frac{P_t}{P_{t+1}}$.

ASSET PRICES

- We define gross inflation rate as

$$\Pi_{t+1} := \frac{P_{t+1}}{P_t}.$$

- We obtain the Fisher equation

$$\frac{1}{1+i_t} = \frac{1}{1+r} E_t \frac{P_t}{P_{t+1}} = \frac{1}{1+r} E_t \frac{1}{\Pi_{t+1}} = \beta E_t \frac{1}{\Pi_{t+1}}.$$

- The usual formulation of the Fisher equation is obtained by linearization:

$$i_t = r + \mathbb{E}_t \pi_{t+1}.$$

FISHER EQUATION

- In general, the Fisher equation is

$$i_t = r_t + \mathbb{E}_t \pi_{t+1}.$$

- This is a no-arbitrage condition: you should be indifferent between two types of investment (real and nominal).

DETERMINACY

- In this endowment economy

$$\frac{1}{1+i_t} = \beta E_t \frac{1}{\Pi_{t+1}}$$

  is the only equilibrium condition – we already used goods market clearing.
- What does it imply for the sequence of price levels $\{P_t\}_{t=0}^{\infty}$?
- In particular, we say that the level of inflation is unique or determinate in equilibrium if:
  1. There is a unique scalar P_0 in equilibrium.
  2. If Π' and Π'' are two sequences that satisfy equilibrium conditions, then $\Pi' = \Pi''$.

DETERMINACY

- The question of determinacy may seem esoteric, but it is important.
- It boils down to the question of whether some policy can control prices.
- In this simple economy, the only available policy is the choice of the nominal interest rate.
- Suppose we have some target for the price level (or inflation rate) – can we achieve it by setting interest rates?

DETERMINACY

- Consider the simplest possible policy: interest rate peg.
- The central bank sets $i_t = i$ for all t – it does not say it will *respond* to something, it is committed to i at all times in all states of the world.
- We have

$$\frac{1}{1+i} = \beta E_t \frac{1}{\Pi_{t+1}}.$$

- This determines $E_t \frac{1}{\Pi_{t+1}}$ – approximately the expected inflation rate.
- But it does not determine Π_{t+1} – the actual inflation rate.
- Agents expect $1/\Pi_t + 1$ to be equal to $1/(1+i)$, but the actual inflation rate is

$$\frac{1}{\Pi_{t+1}} = E_t \frac{1}{\Pi_{t+1}} + \epsilon_{t+1}, \quad E_t \epsilon_{t+1} = 0.$$

DETERMINACY

- What if we have perfect foresight?
- Then it is not a problem, because $\epsilon_{t+1} = 0$.
- For all $t \geq 0$ Π_{t+1} is determined by the peg, but P_0 is not (because Π_0 is not).
- This is because we do not have a Fisher equation for $t = 0$.
- Conclusion: interest rate peg does not ensure price level determinacy.

DETERMINACY

- To simplify the analysis, we will now focus on linearized Fisher equation

$$i_t = r + \mathbb{E}_t \pi_{t+1}.$$

- Consider an interest rate rule

$$i_t = r + \phi \pi_t$$

.

- We call these rules Taylor rules. In general there can be other variables in the rule, and the intercept can be different than r.
- Use the expression for i_t in the Fisher equation to obtain

$$\phi \pi_t = \mathbb{E}_t \pi_{t+1}.$$

DETERMINACY

- We have the difference equation

$$\pi_t = \frac{1}{\phi} \mathbb{E}_t \pi_{t+1}.$$

- We can solve it forward as

$$\pi_t = \lim_{T \to \infty} \phi^{-T} \mathbb{E}_t \pi_{t+T}.$$

- If $\phi > 1$ and $E_t \pi_{t+T}$ does not diverge, then $\pi_t = 0$ for all t.

DETERMINACY

- If $\phi > 1$ and $E_t \pi_{t+T}$ does not diverge, inflation is determinate in all periods, even at $t = 0$.
- π_0 is determined, so P_0 is determined too.
- If the response of the interest rate to inflation is strong enough, $\phi > 0$ then $\pi_t = 0$ for all t is the only bounded solution.
- We call $\phi > 1$ the Taylor principle.
- If $\phi < 1$, then π_t is not determined: there are multiple bounded solutions of the difference equation.
- Where does $E_t \pi_{t+T}$ not diverging come from?

DETERMINACY

- The terminal condition on $E_t \pi_{t+T}$ is not an optimality condition.
- Even if $\phi > 1$, but if the household for whatever reason expects inflation to diverge (i.e., hyperinflation), it will.
- In this model $\phi > 1$ has nothing to do with the logic like "increase real rates to reduce aggregate demand"
- In a way, the central bank makes a threat: if you do not behave, I will make inflation explode. Is it reasonable? Is it how central banks do it?
- See Cochrane (2011) for a discussion of Taylor rules.
- My advice: you might not like the model (it is obviously simple) – but do not apply some outside logic to understand these results!

DETERMINACY

- The usual justification is that when inflation gets out of control, the central bank will switch to something else.
- But then, it is not a Taylor rule with $\phi > 1$ that ensures determinacy!
- What is this other thing that would ensure determinacy?
- For example, the ECB has a monetary pillar, understood as a commitment to switch to a monetary approach to pin down inflation if inflation starts exploding.
- We will soon turn to these monetary approaches.

DETERMINACY

- We used r as the intercept. Suppose r_t is not constant and the central bank follow a more standard Taylor rule:

$$i_t = \bar{r}_t + \pi^* + \phi\left(\pi_t - \pi^*\right),$$

  where $\bar{r}_t$ is the intercept, π^* is the target inflation rate and ϕ is the response to inflation.
- If we use this rule in the Fisher equation, we obtain

$$\pi_t - \pi^* = \frac{1}{\phi}\left[r_t - \bar{r}_t + \left(E_t\left[\pi_{t+1}\right] - \pi^*\right)\right]$$

- What is needed to ensure that $\pi_t = \pi^*$ for all t?

DETERMINACY

- The problem is now harder, because r_t possibly fluctuates.
- In a richer model r_t would depend on the monetary policy itself: monetary policy would affect the real economy.
- To know how, you would need to know how inflation is linked to output (e.g., through the Phillips curve).
- Holden (2024) shows that if you replace $\bar{r}_t$ with r_t in the monetary policy rule, you will restore determinacy – Robust Real Rate Rules (subject to the qualification that $E_t \pi_{t+T}$ does not diverge).
- If there is a unit response to real interest rates, i.e., $\bar{r} = r_t$, then you do not need to know other details of the economy to set inflation at the target. You only need a Fisher equation in the economy and you need to observe r_t somehow.
- This is an extremely powerful result.

DETERMINACY

- Assume there is some money demand in this economy:

$$m_t = p_t + y - \eta i_t,$$

  where $\eta > 0$ is a semielasticity of money demand.
- Use the Fisher equation to obtain

$$m_t = p_t + y - \eta \left[r + \mathbb{E}_t \pi_{t+1} \right].$$

- Let the central bank follow some money growth rule, e.g. $m_t = m_{t-1} + \bar{m}$.
- Is it enough to ensure price level determinacy?

DETERMINACY

- We have a formula

$$(1+\eta)\left(p_t - \bar{m}\cdot t\right) = \eta\left(\mathbb{E}_t\, p_{t+1} - \bar{m}\cdot t\right) - y + p_t + \eta r.$$

  we can solve it forward and there will be a term like $\lim_{T\to\infty} \mathbb{E}_t\, p_{t+T} - \bar{m}\cdot(t+T-1)$.
- This term will be equal to 0 by the household's optimality condition (transversality condition).
- Intuition: households do not want to hold too much or too little money.
- We get

$$p_t = \bar{m}\cdot t + \eta\bar{m} + \frac{1}{1+\eta}\sum_{j=0}^{\infty}\left(\frac{\eta}{1+\eta}\right)^j(\eta r - y)\,.$$

DETERMINACY

- Money growth rule ensures price level determinacy (in this case).
- This is true even if there are shocks u_t to money demand.
- With different microfoundations for money demand, conclusions might be different.
- We can solve for the implied nominal interest rate:

$$i_t = \frac{1}{\eta}\left(p_t - m_t + y\right)$$

- Note it responds to price level. This is an equilibrium outcome.
- If we instead started with a rule that responds to price levels it would be a Wicksellian rule.

BUDGET CONSTRAINT?

- So far we have seen that fiscal policy can affect inflation / price levels by affecting supply / demand of money.
- Is there any other link? In particular, what if the economy is cashless?
- One possibility is that fiscal policy has effects on output and then on inflation through the Phillips curve.
- We will focus on a different link that could be present even in an endowment economy.
- That link will depend on how we interpret the government budget "constraint".

BUDGET CONSTRAINT?

- Example: simple static economy.
- In the morning, bondholders wake up and own B_0 units of zero coupon bonds that promise to pay one unit of account ("dollar") each.
- The government pays bondholders by printing “money".
- In the evening, the government collects taxes $P_1 T_1$. Taxes are paid in "money".
- The world ends aferwards.

BUDGET CONSTRAINT?

- In equilibrium

$$B_0 = P_1 T_1.$$

- “Money" printed up in the morning must be all “soaked" up by taxes at the end of the day.
- B_0 is predetermined. We assume T_1 is exogenous.
- The price level P_1 adjusts to ensure this equality.
- This is an equilibrium condition.
- It looks a bit like a budget constraint, but it is not.

BUDGET CONSTRAINT?

- In the example above we did not require the government to "collect enough taxes to pay its debt".
- This is the Fiscal Theory of the Price Level.
- Consider other possible stories about how P_1 adjusts to ensure $B_0 = P_1 T_1$.
- If P_1 is too low, more money was printed that will be collected in taxes. People will try to get rid of money by buying goods. This will push up the price level.
- “Too much money chasing too few goods".
- Too much government debt relative to fiscal surpluses is like net wealth – people will try to spend more.

FTPL INTRO

- Contrast this with the quantity theory of money.
- In the quantity theory of money, inflation results from more money than needed to settle transations or to satisfy other money demand for other reasons. All money supply matters.
- Here only outside money – government liabilities – matter. It is also irrelevant whether it is debt or currency.
- In the example it is not essential that the government "prints" money: people could pay taxes directly with bonds.

FTPL INTRO

- The government issues the currency and nominal debt that define the price level.
- One possible interpretation nominal government debt is like corporate *equity*: its price adjusts to make a valuation equation hold.
- This does not hold for real or foreign currency government debt.
- This does not hold for personal or corporate debt.

FTPL INTRO

- Note that the government could choose to set taxes T_1 such that $T_1 = \frac{B_0}{P_1}$ for all possible P_1, this means $T_1 = f(P_1)$.
- In this case the price level is indeterminate.
- We call such a fiscal policy passive or Ricardian.
- The point is that the government does not have to follow such a policy. Fiscal policy might be active or non-Ricardian.

AN INTERTEMPORAL MODEL

- We now consider the most basic fully-specified version of the FTPL.
- A representative households maximizes

$$\sum_{t=0}^{\infty} \beta^t u(c_t)$$

  in a complete asset market economy, with restriction $B_t \geq 0$ and $M_t \geq 0$.
- The household has a constant endowment every period, y
- The budget constraint of the household is

$$M_{t-1} + B_{t-1} + P_t y = P_t c_t + P_t s_t + M_t + Q_t B_t$$

  where $Q_t := \frac{1}{1+i_t}$ is the price of bonds in terms of money and s_t is real taxes net of transfers.

AN INTERTEMPORAL MODEL

- Notice that if $Q_t < 1$, then the household does not hold any money, $M_t = 0$.
- If $Q_t = 1$, then money and bonds are perfect substitutes and we can use B_t to denote their sum.
- $Q_t > 1$ not possible.
- Drop M_t from the budget constraint and use market clearing $c_t = y$ to obtain

$$B_{t-1} = P_t s_t + Q_t B_t$$

AN INTERTEMPORAL MODEL

- Household optimality together with the market clearing implies

$$Q_t = \beta \, \mathbb{E}_t \frac{P_t}{P_{t+1}}.$$

- There is also the transversality condition

$$\lim_{T \to \infty} \beta^T \, \mathbb{E}_t \frac{B_{T-1}}{P_T} = 0.$$

- We obtain

$$\frac{B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j s_{t+j}.$$

AN INTERTEMPORAL MODEL

- The government debt valuation equation

$$\frac{B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j s_{t+j}.$$

  determines the price level P_t.
- It does so in every period.
- This looks like a standard asset pricing formula!
- The government debt looks like a stock. Price per share $1/P_t$ times number of shares B_{t-1} equals the present value of dividends $\{s_{t+j}\}$.
- There are problems with this analogy, it is potentially misleading – see Hagedorn (2024).

AN INTERTEMPORAL MODEL

- What if the government changes nominal debt without changing surpluses?
- What if the government changes surpluses without changing nominal debt?
- It is useful to rewrite the equation as

$$\frac{B_t}{P_{t+1}} = \mathbb{E}_{t+1} \sum_{j=0}^{\infty} \beta^j s_{t+j+1}.$$

- Define $\Delta \mathbb{E}_{t+1} := \mathbb{E}_{t+1} - \mathbb{E}_t$ to write

$$\frac{B_t}{P_t} \Delta \mathbb{E}_{t+1} \frac{P_{t+1}}{P_t} = \Delta \mathbb{E}_{t+1} \sum_{j=0}^{\infty} \beta^j s_{t+j+1}.$$

AN INTERTEMPORAL MODEL

- At time $t+1$ both B_t and P_t are predetermined.
- Unexpected inflation is determined by changing expectations of the present value of surpluses.
- Bad fiscal news affect inflation for one period only.
- This is because we assumed one-period debt only.

AN INTERTEMPORAL MODEL

- We can also rewrite the equation as

$$\frac{B_t}{P_t}\mathbb{E}_t\frac{P_{t+1}}{P_t} = \mathbb{E}_t\sum_{j=0}^{\infty}\beta^j s_{t+j+1}.$$

- It is the same as

$$\frac{B_t}{P_t}\frac{1}{1+i_t} = \mathbb{E}_t\sum_{j=0}^{\infty}\beta^j s_{t+j+1}.$$

- The left hand side is the real revenue the government gets from selling debt B_t at the end of period t.
- The price level P_t is already determined at the beginning of period t.
- The revenue from bonds sales is fully determined by the PDV of surpluses.

AN INTERTEMPORAL MODEL

- The government can target nominal interest rates by changing B_t without affecting surpluses.
- By standard arguments the expected inflation rate depends on the nominal interest rate.
- Monetary policy: changing i_t / changing B_t without adjusting surpluses.
- Fiscal policy: changing surpluses without adjusting i_t / B_t.